

very simple set-up. Of course, there is no limit when it comes to a studio mix experience. But the technology supports a Dolby Atmos mix that you can even do through a PC at home, and that's the beauty of the technology. So for the creator, both the formats are available. They can create it on the laptop, headphones, or in a studio.

Q We experienced a bit of Dolby Vision and Dolby Atmos in the studio. How do they work in tandem across different devices?

Sameer: I think there are two parts to the question. One, Dolby Vision and Dolby Atmos. So Dolby Vision is a visual technology from Dolby. It's an HDR technology and one can experience it on different devices. It's available in the cinemas, not in India, but outside of India. You get Dolby Vision experience on your televisions, on your phones, on your PC. So it's a visual technology which gives you better colour, brighter brights and darker darks. So, you know, the brights are 40 times brighter and the darks are ten times darker. So you see a better dynamic range of colours and contrast. Dolby Atmos is an object-based audio technology where the content creator creates the mix by treating sound as objects. It's literally something that will transport you into the experience that will bring you closer to the character and the stories you love. So the combined experience of Dolby Vision and Dolby Atmos is the best possible experience that consumer can get.

Q From what I know, there are three codecs used in Dolby Atmos – Dolby Digital Plus, Dolby TrueHD, and Dolby MAT. Can you delve into their details?

Jayant: Dolby Digital Plus, essentially, is a lossy high compression codec. I think Dolby Digital is the one everyone's familiar with which is now pretty old and has bitrate and compression limitations. Dolby Digital was intrinsically invented for cinema and when it served the purpose of cinema, we stopped innovating briefly. The same thing was adapted for broadcast, pretty much, in the early days. Dolby Digital Plus came about from the streaming era. It has better compression and it has, theoretically, a

larger channel capacity. So Dolby Digital was limited to 5.1. In theory, Dolby Digital Plus can do 13.1. So Dolby Digital Plus, we adapted it because it was pretty much universally in many, many devices. When we launched Atmos, we had to figure out a way of keeping it compatible because of the huge device deployment. So, we retrofitted Atmos with a little bit of clever engineering and the same codec can carry Atmos. But, it is, in essence, a compressed lossy codec for streaming and broadcast. TrueHD and MAT, on the other hand, are very very high quality. TrueHD is lossless completely. There's no lossy compression. Again, TrueHD came out with Blu-ray, pretty much. That was the default lossless codec on many Blu-rays. Not all, but many. And Dolby MAT then became a transport mechanism to go from what we call a source device to a seat device, a set-top box connected to a TV, for example. So, you can decode Atmos in a set top box and then over HDMI 2.x, you can stream uncompressed audio. So, if your TV doesn't have a decoder, you can actually decode Atmos into uncompressed audio and then transport it with metadata using MAT.

The set-top box knows what the TV can render or what the downstream device using EDID and various signalling techniques. Then it will do the render. MAT allows you to move spatial audio into that device and then use metadata, which is the object metadata, so we can transport that across.

Q How is the rendering happening on your end via AI or algorithms, so that it knows what device it's playing on and adjusts accordingly?

Jayant: So in channel-based traditional mixing, stereo for example or 5.1, the mix engineer basically does the render. He or she assigns sound to specific speakers or channels. So if you do a 5.1 mix, you could downmix it, you know, but even the downmix is sort of rigid, it follows

certain rules. You can downmix 5.1 to say stereo or mono, that's it. You can't do fancy configurations like three or four speakers. In an object-based sound format like Atmos, this assignment of sound to the speaker doesn't happen. What happens is that the mix engineer is basically working with a bunch of objects. So, there are certain static sounds assigned by the mix engineer. He or she has a certain speaker configuration which he is monitoring on, and then he just assigns sounds to locations in a room or a 3D space. So, he can place a sound above his head to the left or be just behind him using a joystick kind of a mechanism with a visual display. It shows you where the sound is physically placed as a visual representation. So, how big that sound is, how loud it is and where it is located are all that you need. There's no allocation of any speaker. What you record is

the sound, its location and importance, right? All of this is transmitted with metadata. The consumer may have different types of speaker configurations. So it may be two speakers, a TV, a soundbar, or a home theatre. So the sound system at home knows the speaker configuration and therefore knows how best to render those coordinates. And then it does it. So, in effect, it becomes independent of the speakers. That's the beauty of immersive systems like Atmos.

Q Right now we're living in the era of 5G and the Metaverse. So what are Dolby's thoughts here and what is the plan for the future in these scenarios?

Sameer: The technology should be something that should support where ever the consumer moves. For us, we will continue to innovate. We will continue to work with the ecosystem to deliver experiences where the consumer goes. So, nothing specific on these formats to share. But yeah, we believe that the consumer is something that we will follow. **Q**



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